

Team DeepMinds

Mohit Jain - 612415107

Tejas Kadam - 612415074

Darshana Kulkarni - 612415090

AI Exploration Challenge Meditron

An Open Medical Foundational Model

The future of Medicine powered by LLM

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Why Meditron?

Large language and multimodal models (LLMs and LMMs) have the potential to **revolutionize access to medical knowledge and clinical decision** support. However, current systems are constrained by limited scale, closed-source designs, or poor adaptation to clinical settings. Meditron, an **open and clinically tuned** LLM suite (7B and 70B), addresses these gaps and could **shape the future of medicine**.



Objective

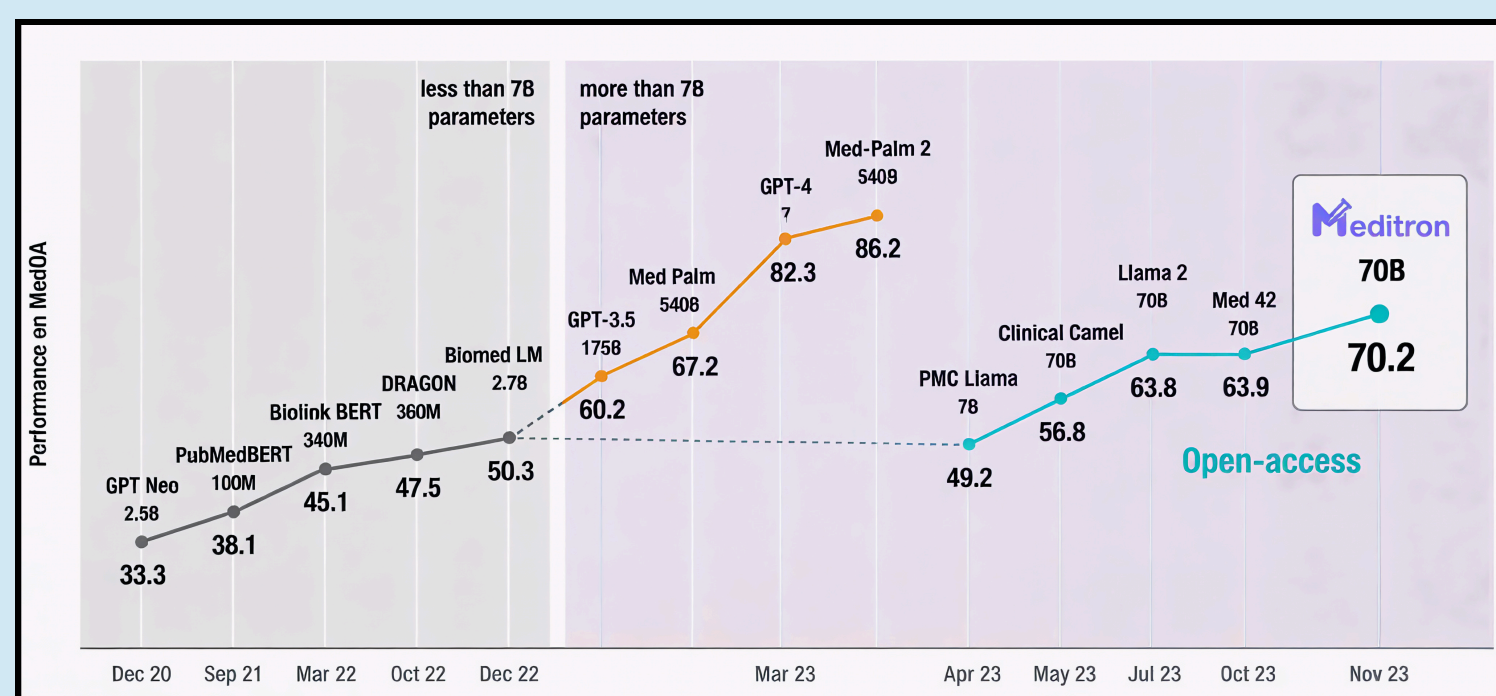


This study evaluates Meditron-7B for inference **accuracy, consistency**, and real-world clinical and research use. Its multimodal extension, MEDITRON-V, could significantly **enhance clinical decision-making** and **future healthcare innovation**.

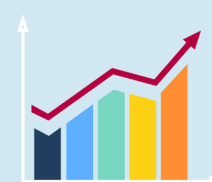


Results

This study demonstrates that **domain specialization** in LLMs is a viable but constrained strategy for edge deployment. Meditron-7B achieves **state-of-the-art medical reasoning** among 7B models, but at the cost of catastrophic degradation in general tasks. At present these models are suitable for **reference lookup** and not diagnosis.



Analysis



Our results show a **strong trade-off** between medical specialization and general reasoning.

- **Medical performance remains strong** despite 4-bit quantization:
- MedMCQA (52%) and MedQA (30% with CoT) stay **close to official baselines**.
- Chain-of-Thought (CoT) is critical for clinical reasoning, improving MedQA from 22% → 30%.
- **General reasoning** collapses:
 - GSM8K (Math): 0-20%
 - HellaSwag (Commonsense): ~25%
 - MMLU-Pro (10 options): 0%

This shows that medical pretraining overwrites general reasoning, but preserves clinical knowledge. CoT acts as "working memory" by forcing the model to generate intermediate states before committing to an answer.

This is especially critical for 7B models, which lack the parameter capacity to perform multi-hop reasoning implicitly.

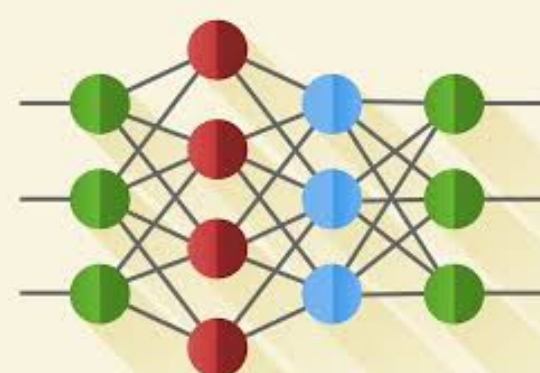
Benchmarking Methodology

Tests were performed on Macbook Air (M1, 2020).

Due to CPU inference speed 50 question samples were taken for each benchmark.

Three paradigms were tested:

1. Zero-Shot Direct
2. Few-Shot
3. Chain-of-Thought



According to official documents

Benchmark	Llama-2-7B	Mistral-7B	Meditron-7B
PubMedQA	56.4%	58.2%	74.4%
MedMCQA	43.1%	48.0%	59.2%
MedQA	35.8%	41.2%	47.9%
Avg. Gain	+14.8%	+11.3%	—

